

FATEHDEEP SINGH WALLA

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Education

- **B. Tech in Mechanical Engineering** | **IIT Delhi** | **2015–2019**
 - ✓ CGPA: 8.25/10
 - **GATE 2026 | Computer Science & Information Technology (CS)**
 - ✓ All India Rank (AIR): 707 among 211,020 candidates
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Work Experience

- **Scientist** | **Indian Space Research Organization (ISRO)** | **2019–2025**
 - ✓ Developed and maintained scientific software for high-performance numerical simulations.
 - ✓ Contributed to the development of PRAVAHA, an in-house Computational Fluid Dynamics solver.
 - ✓ Quantification of Aero and Thermal loads on Launch Vehicle during various phases of flight.
 - ✓ Design and performance evaluation through CFD simulations and wind tunnel data.
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Relevant Knowledge & Skills

- **Computer Science Fundamentals**
 - ✓ Discrete Mathematics | Theory of Computation
 - ✓ Digital Logic | Computer Organization & Architecture
 - ✓ Data Structures & Algorithms
 - ✓ Operating Systems | Computer Networks
 - ✓ Databases | Compiler Design
- **Software Engineering**
 - ✓ C Programming | Systems Programming | Linux | CMake
 - ✓ Java | Object-Oriented Programming | Maven
 - ✓ JUnit | Mockito | Unit Testing | Integration Testing
 - ✓ OOAD | Design Patterns
 - ✓ SQL | JDBC | JPA (Jakarta Persistence)
 - ✓ Concurrent, Parallel & Distributed Programming
 - ✓ Version Control: Git & GitHub
- **Frontend Development**
 - ✓ HTML | CSS | JavaScript | React.js
- **Java Backend Development**
 - ✓ Spring Core (IoC Container & Dependency Injection) | Spring AOP
 - ✓ Spring Boot | Spring MVC | REST API Development
 - ✓ Thymeleaf | Postman | OpenFeign | RestTemplate | WebClient
 - ✓ JdbcTemplate | Spring Data | Spring Transactions

Professional Certification

- **Java Programming and Software Engineering Fundamentals Specialization**
 - ✓ Duke University
- **Object-Oriented Analysis and Design Specialization**
 - ✓ University of Colorado Boulder
- **Parallel, Concurrent, and Distributed Programming in Java Specialization**
 - ✓ Rice University
- **Data Structures and Algorithms Specialization**
 - ✓ University of California San Diego
- **Open Source Software Development, Linux and Git Specialization**
 - ✓ The Linux Foundation
- **Meta Front-End Developer Professional Certificate**
 - ✓ Meta